

Md Mizanur Rahman (Tanim)

IoT & Full-Stack Developer — Edge AI & GenAI Integration

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Summary

IoT and full-stack developer with 5+ years of challenging experience spanning embedded systems, web platforms, and automation with a rare crossover into generative AI integration and local/edge AI. Currently a Developer at Wish and Cook in Portugal, building and maintaining the company's IoT gateway and a custom Modbus protocol stack in Python and C, and integrating generative AI into products and internal workflows. Previously ran an independent development practice in Sweden delivering full-stack e-commerce and ERP automation for multiple clients. Passionate about fault-tolerant software, edge computing, local/edge AI, and AI-driven automation. Although I have never had the chance to work with Elixir (BEAM VM), I am very interested in and optimistic about learning it and exploring its capabilities for building fault-tolerant and highly concurrent systems. Open to remote or hybrid roles across the EU.

Experience

Developer — Wish and Cook, Ílhavo, Portugal (2023 – Present)

- Build and maintain the company's custom Modbus RTU protocol and gateway written in Python and C with MQTT messaging, architected around a fault-tolerant actor model.
- Write custom applications and scripts for hardware testing and validation.
- Built a generative-AI pipeline that converts human-written cooking instructions into a machine-readable format, editable in a Next.js-based cookbook.
- Support the team with data analysis, architecture decisions, and programming challenges.

Independent Developer — Sweden, remote (2018 – 2023)

- Built the Nawras Foods e-commerce platform on Shopify with a custom FastAPI/Python backend that automates product sync and generates offers and invoices in Fortnox ERP.
- Delivered full-stack web and automation projects for multiple clients.

Selected Projects

- Nimai — Telegram bot built with the OpenAI Agent SDK and the KuzuDB embedded graph database (hobby project) (github.com/tanimtbh/nimai)
- Dispenser Fault-Tolerant App — Fault-tolerant dispenser control application built on an actor-model design. This project I made personally, based on a real problem at my company, to showcase my skills and the necessity of the actor model. (Abstract) (github.com/tanimtbh/dispenser_fault_tolerant_app)
- Hands-On Effective Use of LLMs — Internal workshop materials on applying LLMs to real development tasks (github.com/tanimtbh/hands_on_effective_use_of_llm_workshop)
- Custom Modbus RTU (STM32) — STM32 firmware project implementing a custom Modbus RTU protocol. (Feel free to ask for more details)
- Recipe CookBook (Next.js) — Next.js-based cookbook application for converting human-written cooking instructions into a machine-readable format. (Feel free to ask for more details)

Education

- BSc in Electronics and Communication Engineering — Sylhet International University, Sylhet, Bangladesh (2012 – 2016)
- Graduate Coursework in Intelligent Transport Systems and Logistics — Linköping University, Norrköping, Sweden (2018)

Skills

- Programming Languages: Python, C, JavaScript
- Occasionally Learning: Elixir OTP
- AI & Data: Generative AI integration (OpenAI, Anthropic, DeepInfra, OpenRouter, local models), vector DBs (Qdrant, Chroma), graph DBs (KuzuDB, Neo4j), data analysis
- Web & Backend: React, Next.js, FastAPI, REST APIs, Shopify
- IoT & Embedded: ESP32, STM32, Raspberry Pi, Modbus, MQTT, Edge Computing, Edge AI (MobileNet v2, YOLO)
- Tools & DevOps: Git, GitHub, Linux (bare metal)

Languages

- English — Professional working proficiency
- Bengali (Bangla) — Native

Availability

Actively seeking remote or hybrid roles to accommodate Portuguese language studies and integration activities.